MECH 325 - Midterm Book 1
Closed Book

Tuesday, October 26, 2021

Name: Solutions

Student Number: _____

Circle your section

Section 102

Section 101

Signature: _____

Part 1 MC Mark _____ / 12

Part 1 SA Mark _____ / 10

Part 2 LA Mark _____ / 38

Total _____ / 60

Instructions

There are two parts to this exam with different instructions for each. Please read carefully.

Part 1 – Closed-Book (22 marks)

Multiple choice (12 marks: 1 marks per question). Complete all 12 questions by marking your response on the paper.

Short answer (10 marks). Complete all 5 questions by marking your response in this exam booklet.

After you have completed Part 1, hand in your paper in order to begin the Open Book portion. You may not return to Part 1 after you hand it in.

Part 2 – Open-Book (38 marks)

Long answer (38 marks): Complete all parts to the questions for Part 2 by marking your responses on the exam paper. This portion of the exam is open-book and open-notes.

NO COMMUNICATION DEVICES!

Part 1A: Multiple Choice (12 Points)

Answer all 12 questions. Each is worth 1 mark. Choose the single best response – a given question may have more than one choice that is correct but marks will only be given for the *best* answer.

1. **What is the best description of the type of motion between the teeth of two properly meshed worm gears?**
   a) Purely rolling
   b) Purely sliding
   c) Some rolling when teeth engage and disengage, but predominantly sliding
   d) Some sliding when teeth engage and disengage, but predominantly rolling
   e) Approximately equal parts sliding and rolling

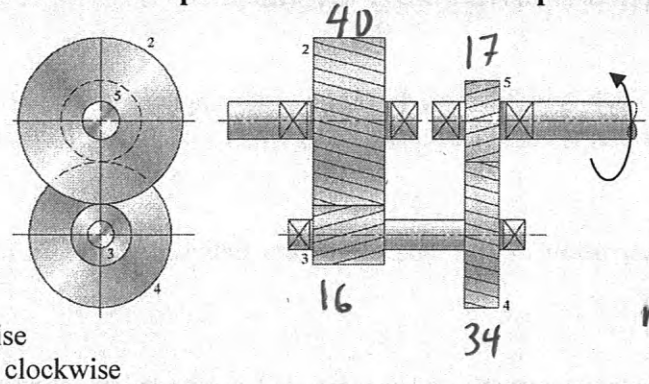
2. **For properly meshed worm gears which of the following statements is FALSE?**
   a) The magnitude of the axial force on the worm is the same as radial force on the gear
   b) The magnitude of the radial force on the worm is the same as the radial force on the gear
   c) The magnitude of the axial force on the worm is the same as the tangential force on the gear
   d) The magnitude of the tangential force on the worm is the same as the axial force on the gear
   e) The magnitude of the friction force on the worm is the same as the frictional force on the gear

3. **Which of the following types of gears is appropriate for high-load transmission of motion between parallel shafts with no axial loading on either shaft?**
   a) Bevel
   b) Hypoid
   c) Worm
   d) Herringbone
   e) Helical

4. **A spur gear has 40 teeth, a 14° pressure angle, and a pitch diameter of 2.0 in. What is the diametral pitch of the gear?**
   a) 0.05
   b) 1.42
   c) 2.86
   d) 7.0
   e) 20.0

$P_D = N/D = 40/2 = 20$

5. **For the gear train shown, gear 5 is the driving pinion Gear 5 has 17 teeth, Gear 4 has 34 teeth. Gear 3 has 16 teeth and Gear 2 has 40 teeth. An 1800 rpm motor is driving the gear box in the Clockwise direction. What is the speed and direction of the output shaft?**



$e = \frac{17}{34} \cdot \frac{16}{40}$

$= \frac{1}{2} \cdot \frac{2}{5} = \frac{1}{5}$

$n_2 = e n_5 = \frac{1}{5}(1800)$

$n_2 = 360 \text{ rpm}$

e is positive so same direction as input

a) 360 rpm clockwise
b) 360 rpm counter clockwise
c) 720 rpm clockwise
d) 720 rpm counter clockwise
e) 300 rpm clockwise

6. **The motor attached to the gearset in Question 4 above is rated at 750 Watt. What is the torque on the output shaft?**

$P_{power} = T\omega \quad 750 = T \, 360/60 \cdot 2\pi$

$T = 750/12\pi = 19.89 \text{ N-m}$

a) 15 N-m
b) 19.9 N-m
c) 23.9 N-m
d) 30 N-m
e) 47.9N-m

7. **The term *"Post-Extreme Horespower"* refers to the following:**
a) The breaking strength of a synchronous belt
b) The power rating for a chain where roller fatique or excessive wear are the modes of failure
c) The power rating for a chain where link fatigue failure is the primary mode of failure
d) The maximum horsepower rating for a chain
e) The smallest degree of lubrication

8. **Which of the following components of belt tension present in V-Belts but normally ignored in flat belts that is the main consideration when calculating belt life?**
a) Initial tension (F_i)
b) Bending tension (F_b)
c) Centrifugal tension (F_c)
d) Tight-side tension (F_1)
e) Slack-side tension (F_2)

9. **Which of the following statements is FALSE for a wormgear drive**
a) Efficiencies of wormgears are less than bevel gears
b) High gear ratios can be achieved using worm gears
c) The efficiency of the wormgear is a function of the friction coefficient and the pressure angle
d) The axial velocity of the worm is the tangential velocity of the gear
e) For wormgears to be *self-locking,* the lead angle must be kept low (no more than 5°)

10. Which of the following statements regarding gears is TRUE?

a) Spur gears can take higher loads than helical gears
b) Helical gears can never be used for right angle drives (shafts 90°)
c) Lower pressure angles allow for higher transverse forces between gears
d) Pinions with only 10 teeth can be used with a rack
e) Miter gears are a form of helical gears

11. For roller chains the term *chordal speed variation* refers to:

a) the speed of engagement of a chain with a sprocket
b) the relationship between the chain speed and the sprocket speed
c) the change in chain speed as the friction in the chain changes
d) the change in chain speed due to the slight rise and fall of the chain as it engages with the sprocket
e) the change in chain speed due to elongation of the chain

12. When specifying the working load of wire rope, which of the statements is FALSE:

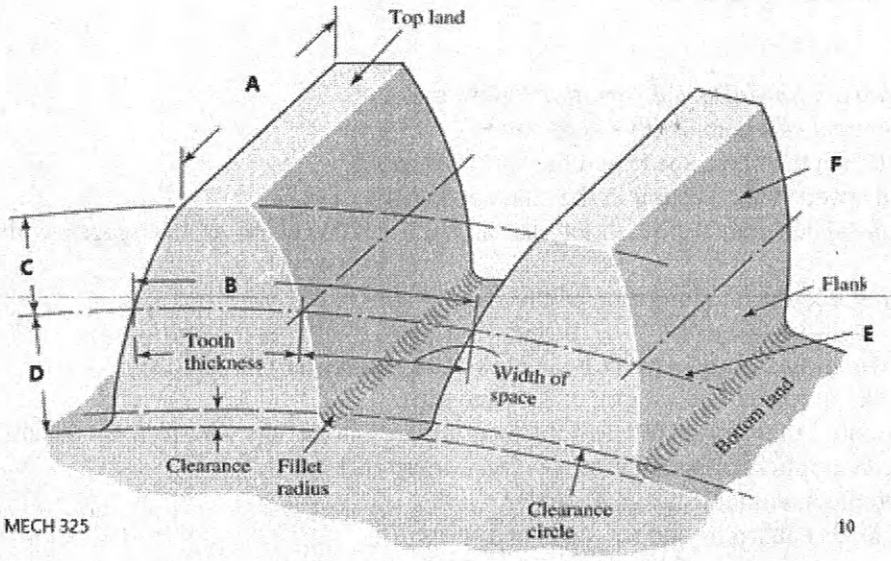
a. one must consider the modulus of elasticity of the wire rope
b. one must take in account the maximum allowable bearing pressure of the wire rope on the sheave
c. one must consider the application in order to specify a safety factor
d. one must determine the maximum sheave diameter
e. one must consider loads caused by sudden stops and starts

13. (Bonus) To convert from horsepower to ft-lbs/sec use the following conversion

a) Divide by 454
b) Divide by 33000
c) Multiply by 550
d) Divide by 550
e) Divide by 32.2

Part 1B – Short-Answer Questions (10 marks)

Answer all questions in the spaces provided.



14. Name the SIX missing terms (A through F) related to spur gears (3 marks)

½ mark each

A = Width (will accept Crown or Depth)
B = Pitch
C = Addendum
D = Dedendum
E = Pitch Circle (will accept Pitch Diameter)
F = Face

15. Which of the above terms are important for stress analysis of the gear teeth? Why (2 marks)?

The pitch, pitch circle, and width are key terms for stress analysis. Both bending & contact stresses are reduced as width, W, increases. A reduction in pitch increases the tooth size which reduces bending stress. Increasing pitch circle will reduce transmitted forces which will reduce the stress (for the same input power)

1 mark

1 mark

16. In gear design, what is meant by interference? How can it be avoided (2 marks)?

Interference relates to gear tooth contact that is not conjugate. It causes the teeth to bind and cause excessive wear.

Interference is avoided by having the teeth engage between the tangent on the base circle formed by the pressure angle. This is achieved by selecting gears with greater number of teeth (avoid pinions with small # of teeth.

1 mk

1 mk

17. Two parallel shafts with 6 inch center distances are to be connected by a pair of gears having a diametrical pitch of 16 teeth/inch. The spur gears have a contact pressure angle of 14.5° and provide a gear reduction of 3.0. Determine the pitch diameter and number of teeth in the pinion gear (3 marks)

$r_p + r_g = 6''$ $\quad P = 16$ teeth/inch $\quad e = 1/3\ (3:1)$

$\therefore\ r_g/r_p = 3 \Rightarrow r_g = 3r_p$

$\therefore\ r_p + 3r_p = 6'' \Rightarrow 4r_p = 6''$ $\quad r_p = 1.5''$ & $r_g = 4.5''$

so $d_p = 3''$ & $d_g = 9''$

pinion $3 \times 16 = 48$ teeth $\quad$ gear $9 \times 16 = 144$ teeth

2 mk right idea

1 mk correct sol'n

MECH 325 - Midterm Book 2
Open Book

Tuesday, October 27, 2020

Name: Solutions

Student Number: _____

Circle your section

Section 101

Section 102

I adhered to the policy of No Communication Devices.

Signature: _____

Instructions

Please read carefully.

No Communication Devices Allowed! Laptops and Tablets are allowed but communication features MUST be disabled.

There are **TWO** Problems to complete.

Part 2 – Open-Book (38 pts)

Long answer: Complete all parts to the question by marking your response in the pages provided. This portion of the exam is open-book and open-notes. **At the end of the exam, return this handout with your answers on the pages.**

This is a multi-part problem. For each part, use the parameter values given. If you cannot complete one part, you may assume some values and move on in order to receive marks for subsequent parts. Clearly indicate your final answer to each part by drawing a box around it. Separate different parts with a line drawn across the page.

Long Answer Problem (38 marks)

Problem 1 (16 Marks)

A chain drive system is required to attach an electric motor to an industrial agitator. The following information relates to the equipment

Driver	AC Motor (High Torque)
Driven Machine	Reciprocating Pump
Service	6 hours/day
Input Speed	700 rpm
Input Power	5 HP
Output Speed (Nominal)	300 rpm
Centerline Distance (Nominal)	30 inches
Design Factor	1.0

2 a) Specify a suitable service factor
4 b) Specify a chain size, and the number of teeth for the sprockets. Select the smaller sprocket to have less than 20 teeth. What is the final velocity ratio?
5 c) Specify the number of chain pitches (links) for the nominal centerline distance
2 d) Specify the type of lubrication regime required for this operation.
1 e) What is the safety factor for your design? Comment on the result

a) Service Factor DriveR is AC Motor (High Torque) (2mk)
DriveN is Reciprocating Pump
From Table 7-17 (MoH) Service Factor = 1.3

b) Input Speed = 700rpm Output Speed = 300rpm (nominal)
Velocity Ratio = 700/300 = 2.33

(2mk) Options to achieve VR ≅ 2.33 [Use Table 17-4 for options]
(i) select 17 tooth sprocket to 40 tooth gear VR = 40/17 = 2.353
(ii) select 18 tooth " " 40 " " VR = 40/18 = 2.22
(iii) select 19 " " " 45 " " VR = 45/19 = 2.368
The best choice is 17 to 40 teeth VR = 2.353 and we have an odd integer sprocket. which is ideal.

Design H.P = (Service Factor) × (Input Power) = 1.3 (5.0)
Design HP = 6.5 HP

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1) M.H: (i) Table 17-14 #40 chain
17 teeth @ 700 rpm gives $H_A = 5.16$ HP < 6.5 HP

(ii) Table 17-15 #50 chain
17 teeth @ 600 rpm gives $H_A = 14.88$ HP
17 teeth @ 800 rpm gives $H_A = 19.62$ HP
∴ 17 teeth @ 700 rpm $H_A = 17.25$ HP $\left(\frac{1}{2}(14.88 + 19.62)\right)$

Select Single Strand #60 chain

2 mks

2) Shigley: Design HP = 6.5 HP
Table 17-20
17 teeth @ 700 rpm $H_{Tab} = 7.73$ HP for #50 chain
$K_1 = 1.0$ for 17 teeth
$K_2 = 1.0$ for single strand
$H_A = K_1 K_2 H_{Tab} = (1.0)(1.0)(7.73) = 7.73$ HP

3 mks

Final Selection: #50 Chain (Single Strand) Sprocket = 17 teeth & 40 teeth
Final VR = 2.353
Output rpm = 297.5 rpm

c) Nominal Center line Distance CD = 30 inches

$$L/P = \frac{2C}{P} + \frac{N_1 + N_2}{2} + \frac{(N_2 - N_1)^2}{4\pi^2 C/P}$$

Shigley Eq'n 17-34
M.H Eq'n 7-18

3 mks

For #50 chain $P = 5/8'' = 0.625$ inches
$N_1 = 17$ $N_2 = 40$ & $C = CD = 30$ inches

2 mks

$$L/P = \frac{2(30)}{0.625} + \frac{17+40}{2} + \frac{(40-17)^2}{4\pi^2(30/0.625)} = 96 + 28.5 + 0.28$$

$L/P = 124.8$ links

Select Even Number Links for 17 tooth Sprocket = 124 links

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d) Lubrication Regime

MoH Table 7-15 Type B

(2 pts) Shigley Table 17-20 Type B

Type B is bath or disk lubrication

e) Safety Factor $n_{SF} = \frac{H_{allowable}}{H_{design}}$

1) MoH $n_{S,F} = 17.25/6.5$

(2 pts) $n_{S,F} = 2.65$ (Perhaps too high)

2) Shigley $n_{S,F} = 7.33/6.5$

$n_{S.F} = 1.19$

Both $n_{S,F} > 1.0$ so design is considered safe.

Alternative Sol'n from MoH

5) #40 chain has $H_{Tab} = 5.16$ @ 700 rpm

Select two strands so $K_2 = 1.7$ & $K_1 = 1.0$ (17 teeth)

$H_{All} = K_1 K_2 H_{Tab} = (1.7)(1.0)(5.16) = 8.78$ HP

e) $n_{SF} = 8.78/6.5 = 1.35$ (reasonable)

d) Type B

c) $L/p = \frac{2C}{p} + \frac{(N_1+N_2)}{2} + \frac{(N_2-N_1)^2}{4\pi^2 C/p}$ #40 chain $p = 0.500$ inch

$L/p = \frac{2(30)}{0.5} + \frac{(17+40)}{2} + \frac{(40-17)^2}{4\pi^2 30/0.5} = 120 + 28.5 + 0.22 = 148.7$

Select 148 links

Problem 2 (22 Points)

Part 1 of 3

(8 points) Do an analysis on the following full depth involute **SPUR** gears:

Note: you can choose imperial or metric units

Driver	AC Motor (High Torque)
Driven Machine	Agitator (Liquid)
Service	4 hours/day
Input Speed	1750 rpm
Input Power	5 HP (3.75 kW)
Pressure Angle	20°
Diametrical Pitch	12 teeth/inch (module = 2 mm.)
Pinion	16 teeth
Gear	51 teeth

Determine the following:

a) Gear Ratio (Velocity Ratio)
b) Pitch diameter for pinion and gear
c) Pitch line speed
d) Tangential and radial forces (W_t and W_r) for the gears

a) Gear ratio $GR = VR = N_G / N_P = 51/16 = \boxed{3.188}$ (1 mk)

b) Pitch Diameter = No. of teeth / Diametrical Pitch

$\boxed{d_{pinion} = 16/12 = 1.333 \text{ inches}}$
$\boxed{d_{gear} = 51/12 = 4.250 \text{ inches}}$ (2 mk)

c) Pitch Line Speed = Tangential Velocity $= \omega R$

$\omega = 1750 \text{ rpm} / 60 \text{ sec/min} \times 2\pi$

$R = d_{pinion}/2$

$\therefore \text{speed} = (1750/60) \times 2\pi \left(1.333/2\right) = 122.17 \text{ in/sec}$

$\boxed{\text{line speed} = 610 \text{ fpm}}$ (2 mk)

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d) Power = Torque × Rotational Speed

$= T\omega$

$\therefore T = \text{Power}/\omega = 63000\, P/n$ [lb in] [Mott eq'n 9-10]

P = HP n = rpm

Power = 5 HP n = 1750 rpm

$T = 63000(5)/1750 = 180$ lb.in

$T = W_t(\text{radius}) = W_T(d_p/2)$

$W_t = T/(d_p/2) = 180 \text{ lb.in}/(1.333 = d_p/2)$

$W_t = 270.1$ lbs

Radial Load $W_r = W_t \tan\phi$ [Eq'n 9-11 Mott]

ϕ = Pressure Angle = 20°

$W_r = (270.1)(\tan 20^\circ)$

$W_r = 98.3$ lbs

3 mk

∴

$W_t = 270.1$ lbf

$W_R = 98.3$ lbf

Part 2 of 3

(10 Points) Determine the AGMA **contact** stress number for the 16 tooth **pinion ONLY.**

Given:

Face Width	1.00 inch (25 mm.)
Pressure Angle	20°
Gear Quality	$A_v = 7$ ($Q = 10$)
Material for Pinion & Gear	Steel
Size Factor	$K_s = 1.0$
Rim thickness factor	$K_B = 1.0$
Overload Factor	$K_o = 1.25$ (uniform power source and light shock)
Elastic Coefficient	$C_p = ?$
Dynamic factor	$K_v = ?$
Load Distribution factor	$K_m = ?$ (For Shigley Text: $C_{mc} = C_{pm} = C_e = 1.0$)
Geometry factor	$I = ?$
Contact stress number	$s_c = ?$

AGMA Contact Stress Equation

$$s_c = \sigma_c = C_p \sqrt{\frac{W_t}{F D_p I} \cdot K_o \cdot K_s \cdot K_m \cdot K_v}$$

[M of H Eq'n 9-21 Shigley Eq'n 14-16]

Elastic Coefficient: Use Table 9-7 (M of H) or Table 14-8 (Shigley)

2 mk

$C_p = 2300 \sqrt{psi}$ for steel on steel

Geometry Factor

(i) M of H: Use Figure 9-17a) $\phi = 20°$ $N_p = 16$ $m_G = 3.188 \approx 3.2$

$I = 0.098$

(ii) Shigley: Use equation 14-23

$$I = \frac{\cos\phi \sin\phi}{2 m_N} \frac{m_G}{m_G + 1}$$

$m_N = 1.0$ for Spur Gears

$m_G = 3.188$ $\phi = 20°$

$$I = \frac{(\cos 20)(\sin 20)}{2} \frac{3.188}{(3.188 + 1)}$$

2 mk

$I = 0.122$

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Dynamic Factor K_v

For Pitch Line Speed = 610 fpm

Gear Quality $A_v = 7$ $Q = 10$

From Figure 9-16a) (M&H) or Figure 14-9 (Shigley)

$$\boxed{K_v \doteq 1.12}$$

or Shigley Eq'n 14-27

$$K_v = \left(\frac{A + \sqrt{V}}{A}\right)^B \qquad A = 50 + 56(1 - B)$$

$$B = 0.25(12 - Q_v)^{2/3}$$

(2nd)

$$\therefore B = 0.25(12 - 10)^{2/3} = 0.25(2)^{0.667} = 0.397$$

$$\text{so } A = 50 + 56(1 - 0.397) = 83.78$$

$$K_v = \left(\frac{83.78 + \sqrt{610}}{83.78}\right)^{0.397} \qquad \boxed{K_v = 1.11}$$

Load Distribution Factor K_m

$$K_m = 1.0 + C_{pf} + C_{ma} \quad \text{(M\&H Eq'n 9-17)}$$

C_{pf} = pinion proportion factor

$$C_{pf} = \frac{F}{10 D_p} - 0.025 \quad \text{For } F \le 1.0 \quad F = 1.0 \text{ inch}$$

$$C_{pf} = \frac{1.0}{10(1.333)} - 0.025 = 0.05$$

Using Fig 9-12 @ $F = 1$ inch & $F/D_p = 1/1.333 = 0.75$

$$C_{pf} \simeq 0.4$$

C_{ma} = mesh alignment factor

*Assume Commercial Quality Gearbox *

$$C_{ma} = 0.127 + 0.0158F - 1.093 \times 10^{-4} F^2$$

$$C_{ma} = 0.127 + 0.0158 - 1.093 \times 10^{-4} \text{ @ } F = 1 \text{ inch} \qquad C_{ma} = 0.14$$

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From Figure 9-13 @ F=1 inch & Commercial Gearing

$C_{ma} = 0.15$

Shigley $K_m = C_{mf} = 1 + C_{mc}(C_{pf}C_{pm} + C_{ma}C_e)$

Given $C_{mc} = C_{pm} = C_e = 1.0$

$K_m = 1 + C_{pf} + C_{ma}$

Same eq'n as above $C_{pf} = 0.05$ $C_{ma} = 0.14$

$K_m = 1 + C_{pf} + C_{ma}$

$= 1 + 0.05 + 0.14$

(2 mk)

$\boxed{K_m = 1.19}$

Hence

$$\sigma_c = S_c = C_p\sqrt{\frac{W_t}{FD_pI} \cdot K_o + K_s \cdot K_m \cdot K_v}$$

with $K_o = 1.25$ and $W_t = 270.1$ lbf $D_p = 1.333$ in $F = 1.000$ inch

$K_s = 1.0$

Mott $I = 0.098$ $S_c = 2300\sqrt{\frac{270.1}{(1.000)(1.333)(I=0.098)} \cdot (1.25)(1.0)(1.19)(1.12)}$

$K_v = 1.12$

$\boxed{\text{Mott} \quad S_c = 134{,}989 \text{ psi} = 135 \text{ ksi}}$

(2 mk)

Shigley $I = 1.22$ $K_v = 1.11$

$S_c = 2300\sqrt{\frac{270.1}{(1.000)(1.333)(0.122)} (1.25)(1.0)(1.19)(1.11)}$

$\boxed{\text{Shigley} \quad S_c = 120{,}443 \text{ psi} = 120 \text{ ksi}}$

Note: Mott is more conservative

Part 3 of 3

(4 points) Select a suitable through hardened steel for the **pinion**. Specify the alloy and heat treatment.

The gear is expected to operate for 10^7 cycles. You can assume reliability factor and temperature factors are 1.0. You can also assume the Hardness-ratio factor, $C_H = 1.0$

What is your safety factor for contact stress? Comment on the result

Would you recommend flame or induction hardening for this gear? Explain.

Given $K_R = K_T = 1.0$ $C_H = 1.0$

For 10^7 cycles Stress Cycle Factor $Z_N = 1.0$

[Figure 9-22 MoH or Fig. 14-15 Shigley] (1mk)

For through-hardened steel teeth use Fig. 9-19 [MoH] or Fig 14-5 [Shigley]

Set $S_{ac} = S_c = 135$ ksi [MoH sol'n]

$S_{ac} = 0.322HB + 29.10$ ksi Grade 1

$135 = 0.322HB + 29.10 \Rightarrow HB = 329$ Brinell Hardness

Typical Gear Materials 1045, 4140 4340 4150 4350 Table 9-3

For HB > 329 one can select alloys such as: (2mk)

4140 OQT 1000	HB = 341	s.f = 341/329 = 1.04	
4140 OQT 700	HB = 461	s.f = 1.40	
4150 OQT 1000	HB = 401	s.f. 1.21 good	
4340 OQT 700	HB = 461	s.f = 1.40	

Select 4150 OQT 1000 s.f. = 1.21 which is below the AGMA criteria of s.f < 1.5

Shigley $S_{ac} = S_c = 120$ ksi [Shigley sol'n]

$120 = 0.322HB + 29.10 \Rightarrow HB = 282$

Select 4140 OQT 1000 HB = 341 s.f: 341/282 = 1.21

This is also acceptable s.f for AGMA

For flame or induction hardened

Again use 1045 4140 4340 alloys [Table 9-8]
4150 4350

This is preferred to through hardening as the surface gives the required strength against contact stress while the core remain ductile to prevent failure.

Imk

Through hardening is not recommended for gears requiring HB > 400 due to inconsistencies in performance. For requirements of HB > 400 use case hardening techniques such as flame or induction hardening. Hence the 4150 OQT 1000 with HB = 401 is right on the border and should be replaced with a surface hardened version of the gear.